

IMPLEMENTATION FOR RESULT NORMALIZATION DATABASES

ASSIGNMENT FOR THE DATABASE



Lecturer :

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YEAR 2026

CASE 1

No	Nomor Mahasiswa	Nama	Hasil Ujian			
			Ke 1	Ke 2	Ke 3	Ke 4
1	01001	Ali	50	60		
2	01002	Budi	80			
3	01003	Budi	50	50	70	
4	01004	Carli	40	50		
5	01005	Dandi	59	41	42	43

```

Session Actions Edit View Help
rentsaissy@localhost:~$ sudo mysql
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 31
Server version: 11.8.6-MariaDB-6 from Debian -- Please help get to 10k stars at https://github.com/MariaDB/Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> Create DATABASE kampus;
Query OK, 1 row affected (0.017 sec)

MariaDB [(none)]> USE kampus;
Database changed

MariaDB [kampus]> CREATE TABLE mahasiswa (
  → nim VARCHAR(5) PRIMARY KEY,
  → nama VARCHAR(50)
  → );
Query OK, 0 rows affected (0.032 sec)

MariaDB [kampus]> INSERT INTO mahasiswa VALUES
  → ('01001','Ali'),
  → ('01002','Budi'),
  → ('01003','Budi'),
  → ('01004','Carli'),
  → ('01005','Dandi');
Query OK, 5 rows affected (0.019 sec)
Records: 5 Duplicates: 0 Warnings: 0

MariaDB [kampus]> CREATE TABLE nilai (
  → id_nilai INT AUTO_INCREMENT PRIMARY KEY,
  → nim VARCHAR(5),
  → ujian_ke INT,
  → nilai INT,
  → FOREIGN KEY (nim) REFERENCES mahasiswa(nim)
  → );
Query OK, 0 rows affected (0.038 sec)

MariaDB [kampus]> INSERT INTO nilai (nim, ujian_ke, nilai) VALUES
  → ('01001',1,50),
  → ('01001',2,60),
  → ('01002',1,80),
  → ('01003',1,50),
  → ('01003',2,50),
  → ('01003',3,70),
  → ('01004',1,40),
  → ('01004',2,50),
  → ('01005',1,59),
  → ('01005',2,41),
  → ('01005',3,42),
  → ('01005',4,43);
Query OK, 12 rows affected (0.018 sec)
Records: 12 Duplicates: 0 Warnings: 0

```

CASE 2

PT. SAMPLE SPECIAL

Jl. HR Rasuna Said No. 343
Kuningan, Setia Budi 12350, Indonesia.
Telp. : 021 5678 7768 Email : sample@special.com
Fax : 021 4564 7569 www.samplespecial.com

INVOICE CUSTOMER - ACCOUNT

SOLD TO :
SAMPLE INDONESIA, PT.

DELIVERED TO :
SAMPLE INDONESIA, PT.
JL. IMAM BONJOL NO.60 JAKARTA PUSAT 10221
INDONESIA

Up. Mr. Dimas Prasetyo

INVOICE NO : 00000145
DATE : 11 February 2013
P/O NO : 00000234
TERMS : Cash/Tunai
PAGE NUMBER : 1

Item	Description	Qty	Unit Price	Amount
1. 1235-5677	RING	6 Pcs	19.04	114.24
2. 7655-8766	PTFE	6 Pcs	325.86	1,955.16
3. 345-9876	VALVE	1 Pcs	672.08	672.08
4. 2567-9777	BALL	12 Pcs	37.72	452.64

E & O.E.

Sub Total	3,194.12
Less : Discount	0.00
P.P.N 10%	319.41
TOTAL	0.00

IMPORTANT NOTES :
IN CASE OF ANY DISCREPANCIES, PLEASE NOTIFY US WITHIN SEVEN (7) DAYS UPON RECEIVING OF THIS INVOICE
INTEREST WILL BE CHARGED AT 1.5% PER MONTH ON ALL OVERDUE ACCOUNT
NO CLAIMS WILL BE ENTERTAINED AFTER 3 WORKING DAYS FROM THE DATE OF DELIVERY
N.E.G. - NEAREST EQUIVALENT GRADE WITHOUT PREJUDICE FOR REFERENCE ONLY
ALL CHEQUES SHOULD BE CROSSED AND MADE PAYABLE TO
PERMATA BANK CEBANG : POLIM, JAKARTA
IDR 7 1111 7156
USD 7 0134 6789

for PT. SAMPLE SPECIAL

RIHANA
Authorised Signature

```
rentsaisy@localhost: ~  
Session Actions Edit View Help  
MariaDB [kampus]> CREATE DATABASE penjualan;  
Query OK, 1 row affected (0.001 sec)  
MariaDB [kampus]> USE penjualan;  
Database changed  
MariaDB [penjualan]> CREATE TABLE customer (  
→ id_customer INT AUTO_INCREMENT PRIMARY KEY,  
→ nama_customer VARCHAR(100),  
→ alamat TEXT  
→ );  
Query OK, 0 rows affected (0.033 sec)  
MariaDB [penjualan]> CREATE TABLE invoice (  
→ no_invoice VARCHAR(20) PRIMARY KEY,  
→ tanggal DATE,  
→ id_customer INT,  
→ terms VARCHAR(50),  
→ subtotal DECIMAL(10,2),  
→ ppn DECIMAL(10,2),  
→ total DECIMAL(10,2),  
→ FOREIGN KEY (id_customer) REFERENCES customer(id_customer)  
→ );  
Query OK, 0 rows affected (0.031 sec)  
MariaDB [penjualan]> CREATE TABLE barang (  
→ id_barang VARCHAR(20) PRIMARY KEY,  
→ nama_barang VARCHAR(100),  
→ harga DECIMAL(10,2)  
→ );  
Query OK, 0 rows affected (0.030 sec)  
MariaDB [penjualan]> CREATE TABLE detail_invoice (  
→ id_detail INT AUTO_INCREMENT PRIMARY KEY,  
→ no_invoice VARCHAR(20),  
→ id_barang VARCHAR(20),  
→ qty INT,  
→ amount DECIMAL(10,2),  
→ FOREIGN KEY (no_invoice) REFERENCES invoice(no_invoice),  
→ FOREIGN KEY (id_barang) REFERENCES barang(id_barang)  
→ );  
Query OK, 0 rows affected (0.032 sec)
```

```
rentsaisy@localhost: ~  
Session Actions Edit View Help  
MariaDB [penjualan]> SHOW TABLES;  
+-----+  
| Tables_in_penjualan |  
+-----+  
| barang               |  
| customer             |  
| detail_invoice       |  
| invoice              |  
+-----+  
4 rows in set (0.001 sec)  
MariaDB [penjualan]> SELECT * FROM customer;  
Empty set (0.001 sec)  
MariaDB [penjualan]> INSERT INTO customer (nama_customer, alamat) VALUES  
→ ('SAMPLE INDONESIA, PT.', 'Jl. IMAM BONJOL NO.60 JAKARTA PUSAT 10221'),  
→ ('MAJU JAYA PT', 'Bandung'),  
→ ('SUKSES MAKMUR PT', 'Surabaya'),  
→ ('ABADI SENTOSA PT', 'Medan'),  
→ ('GLOBAL TEKNIK PT', 'Bekasi');  
Query OK, 5 rows affected (0.016 sec)  
Records: 5 Duplicates: 0 Warnings: 0  
MariaDB [penjualan]> INSERT INTO invoice  
→ (no_invoice, tanggal, id_customer, terms, subtotal, ppn, total)  
→ VALUES  
→ ('00000145', '2013-02-11', 1, 'Cash/Tunai', 3194.12, 319.41, 3513.53),  
→ ('00000146', '2013-02-12', 2, 'Transfer', 2000.00, 200.00, 2200.00),  
→ ('00000147', '2013-02-13', 3, 'Cash', 1500.00, 150.00, 1650.00),  
→ ('00000148', '2013-02-14', 4, 'Tempo 7 Hari', 5000.00, 500.00, 5500.00),  
→ ('00000149', '2013-02-15', 5, 'Transfer', 1000.00, 100.00, 1100.00);  
Query OK, 5 rows affected (0.019 sec)  
Records: 5 Duplicates: 0 Warnings: 0  
MariaDB [penjualan]> INSERT INTO barang VALUES  
→ ('1235-5677', 'RING', 19.04),  
→ ('7655-8766', 'PTFE', 325.86),  
→ ('345-9876', 'VALVE', 672.08),  
→ ('2567-9777', 'BALL', 37.72),  
→ ('7788-1122', 'PACKING', 55.00);  
Query OK, 5 rows affected (0.018 sec)  
Records: 5 Duplicates: 0 Warnings: 0
```

```
rentsaisy@localhost: ~
Session Actions Edit View Help

MariaDB [penjualan]> INSERT INTO detail_invoice
→ (no_invoice, id_barang, qty, amount)
→ VALUES
→ ('00000145', '1235-5677', 6, 114.24),
→ ('00000145', '7655-8766', 6, 1955.16),
→ ('00000145', '345-9876', 1, 672.08),
→ ('00000145', '2567-9777', 12, 452.64),
→ ('00000146', '7788-1122', 10, 550.00);
Query OK, 5 rows affected (0.022 sec)
Records: 5 Duplicates: 0 Warnings: 0

MariaDB [penjualan]> SELECT * FROM customer;
+----+-----+-----+
| id_customer | nama_customer | alamat |
+----+-----+-----+
| 1 | SAMPLE INDONESIA, PT. | JL. IMAM BONJOL NO.60 JAKARTA PUSAT 10221 |
| 2 | MAJU JAYA PT | Bandung |
| 3 | SUKSES MAKMUR PT | Surabaya |
| 4 | ABADI SENTOSA PT | Medan |
| 5 | GLOBAL TEKNIK PT | Bekasi |
+----+-----+-----+
5 rows in set (0.001 sec)

MariaDB [penjualan]> SELECT * FROM invoice;
+----+-----+-----+-----+-----+-----+
| no_invoice | tanggal | id_customer | terms | subtotal | ppn | total |
+----+-----+-----+-----+-----+-----+
| 00000145 | 2013-02-11 | 1 | Cash/Tunai | 3194.12 | 319.41 | 3513.53 |
| 00000146 | 2013-02-12 | 2 | Transfer | 2000.00 | 200.00 | 2200.00 |
| 00000147 | 2013-02-13 | 3 | Cash | 1500.00 | 150.00 | 1650.00 |
| 00000148 | 2013-02-14 | 4 | Tempo 7 Hari | 5000.00 | 500.00 | 5500.00 |
| 00000149 | 2013-02-15 | 5 | Transfer | 1000.00 | 100.00 | 1100.00 |
+----+-----+-----+-----+-----+-----+
5 rows in set (0.000 sec)

MariaDB [penjualan]> SELECT * FROM barang;
+----+-----+-----+
| id_barang | nama_barang | harga |
+----+-----+-----+
| 1235-5677 | RING | 19.04 |
| 2567-9777 | BALL | 37.72 |
| 345-9876 | VALVE | 672.08 |
+----+-----+-----+
```

```
rentsaisy@localhost: ~
Session Actions Edit View Help

+----+-----+-----+
| 3 | SUKSES MAKMUR PT | Surabaya |
| 4 | ABADI SENTOSA PT | Medan |
| 5 | GLOBAL TEKNIK PT | Bekasi |
+----+-----+-----+
5 rows in set (0.001 sec)

MariaDB [penjualan]> SELECT * FROM invoice;
+----+-----+-----+-----+-----+-----+
| no_invoice | tanggal | id_customer | terms | subtotal | ppn | total |
+----+-----+-----+-----+-----+-----+
| 00000145 | 2013-02-11 | 1 | Cash/Tunai | 3194.12 | 319.41 | 3513.53 |
| 00000146 | 2013-02-12 | 2 | Transfer | 2000.00 | 200.00 | 2200.00 |
| 00000147 | 2013-02-13 | 3 | Cash | 1500.00 | 150.00 | 1650.00 |
| 00000148 | 2013-02-14 | 4 | Tempo 7 Hari | 5000.00 | 500.00 | 5500.00 |
| 00000149 | 2013-02-15 | 5 | Transfer | 1000.00 | 100.00 | 1100.00 |
+----+-----+-----+-----+-----+-----+
5 rows in set (0.000 sec)

MariaDB [penjualan]> SELECT * FROM barang;
+----+-----+-----+
| id_barang | nama_barang | harga |
+----+-----+-----+
| 1235-5677 | RING | 19.04 |
| 2567-9777 | BALL | 37.72 |
| 345-9876 | VALVE | 672.08 |
| 7655-8766 | PTFE | 325.86 |
| 7788-1122 | PACKING | 55.00 |
+----+-----+-----+
5 rows in set (0.000 sec)

MariaDB [penjualan]> SELECT * FROM detail_invoice;
+----+-----+-----+-----+-----+
| id_detail | no_invoice | id_barang | qty | amount |
+----+-----+-----+-----+-----+
| 1 | 00000145 | 1235-5677 | 6 | 114.24 |
| 2 | 00000145 | 7655-8766 | 6 | 1955.16 |
| 3 | 00000145 | 345-9876 | 1 | 672.08 |
| 4 | 00000145 | 2567-9777 | 12 | 452.64 |
| 5 | 00000146 | 7788-1122 | 10 | 550.00 |
+----+-----+-----+-----+-----+
5 rows in set (0.001 sec)

MariaDB [penjualan]>
```